

3.复数类和运算符重载

分数类构造函数

C++

```
1 //本题主要考察-构造函数的定义，根据后缀，和程序输出完成分数类的定义，
2 //包括：
3 //构造函数
4 //show函数
5 //to_double() 函数 ， 即把分数转变为小数
6 //还有
7 //void ifLessSwap(Fraction &a,Fraction &b); 即如果分数a比b小，则交换a和b；否则不交换。
8 //输入：
9 //2 3 1
10 //输出：
11 //0.666667
12 //2/3
13 //1/1
14 //1/1
15 //2/3
16 #include <iostream>
17 using namespace std;
18 class Fraction {
19 private:
20     int numerator;    // 分子
21     int denominator; // 分母
22 public:
23     // 构造函数，初始化分子和分母
24     Fraction(int num, int denom) : numerator(num), denominator(denom) {}
25     // 显示分数的值
26     void Show() {
27         cout << numerator << "/" << denominator << endl;
28     }
29     // 将分数转换为小数形式
30     double to_double() {
31         return static_cast<double>(numerator) / denominator;
32     }
33     // 如果分数 a 比分数 b 小，则交换它们
34     friend void ifLessSwap(Fraction &a, Fraction &b);
35 };
36 // 如果分数 a 比分数 b 小，则交换它们
37 void ifLessSwap(Fraction &a, Fraction &b) {
38     if (a.to_double() < b.to_double()) {
39         swap(a.numerator, b.numerator);
40         swap(a.denominator, b.denominator);
41     }
42 int main() {
43     int num, denom;
44     cin >> num >> denom;
45     Fraction f1(num, denom);
46     Fraction f2(1, 1);
47     cout << f1.to_double() << endl;
48     f1.Show();
49     f2.Show();
50     ifLessSwap(f1, f2);
51     f1.Show();
52     f2.Show();
53     return 0;}
```

复数类输出

```

2  #include<iostream>
3  using namespace std;
4  class Cmycomplex{
5  public:
6      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;};
7      void Set(double xx,double yy){x=xx;y=yy;};
8      void Show(){cout<<"("<<x<<"+"<<y<<"i");};
9      void Add(const Cmycomplex &z){x=x+z.x; y=y+z.y;}
10 private:
11     double x,y;
12 };
13 //StudybarCommentBegin
14 int main()
15 {
16     Cmycomplex  z1(2,3),z2,z3(3);
17     z1.Add(z3);
18     z1.Show();
19 }
    //StudybarCommentEnd

```

复数相加带输入

```

1  #include<iostream>
2  using namespace std;
3  class Cmycomplex{
4  public:
5      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;};
6      void Set(double xx,double yy){x=xx;y=yy;};
7      void Show(){cout<<"("<<x<<"+"<<y<<"i");};
8      Cmycomplex operator+(Cmycomplex &z) //返回值类型是Cmycomplex
9      {
10         Cmycomplex t;
11         t.x=x+z.x;
12         t.y=y+z.y;
13         return t;
14     }
15 private:
16     double x,y;
17 };
18 //StudybarCommentBegin
19 int main()
20 {
21     Cmycomplex  z1(2,3),z2,z3(3);
22     z2=z1+z3;
23     z2.Show();
24     double  x,y;
25     cin>>x>>y;
26     z2.Set(x,y);
27     z3=z1+z2;
28     cout<<"\n";
29     z3.Show();}
30 //StudybarCommentEnd

```

C++

复数相加 add

```

1  #include<iostream>
2  using namespace std;
3  class Cmycomplex{
4  public:
5      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;};
6      void Set(double xx,double yy){x=xx;y=yy;};
7      void Show(){cout<<"("<<x<<"+"<<y<<"i");};
8      Cmycomplex Add(Cmycomplex &z) //返回值类型是Cmycomplex
9      {
10         Cmycomplex t;
11         t.x=x+z.x;
12         t.y=y+z.y;
13         return t;

```

C++

```

14     }
15 private:
16     double x,y;
17 };
18 //StudybarCommentBegin
19 int main()
20 {
21     Cmycomplex  z1(2,3),z2,z3(3);
22     z2=z1.Add(z3);
23     z2.Show();
24 }
25 //StudybarCommentEnd

```

复数 newadd

```

1  #include<iostream>
2  using namespace std;
3  class Cmycomplex{
4  public:
5      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;};
6      void Set(double xx,double yy){x=xx;y=yy;};
7      void Show(){cout<<"("<<x<<"+"<<y<<"i)";};
8      void Add(const Cmycomplex &z){x=x+z.x; y=y+z.y;}//没有返回值对比newadd
9      Cmycomplex Newadd(Cmycomplex &z)//返回值类型是Cmycomplex
10     {
11         Cmycomplex t;
12         t.x=x+z.x;
13         t.y=y+z.y;
14         return t;
15     }
16 private:
17     double x,y;
18 };
19 //StudybarCommentBegin
20 int main()
21 {
22     Cmycomplex  z1(2,3),z2,z3(3);
23     //z2=z1.Add(z3);
24     z1.Show();
25     z3=z1.Newadd(z2);
26     z3.Show();
27 }
28 //StudybarCommentEnd

```

C++

复数加减乘除等于不等于

```

1  #include<iostream>
2  #include<iomanip>
3  using namespace std;
4  class Cmycomplex{
5  public:
6      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;}
7      void Set(double xx,double yy){x=xx;y=yy;}
8      void Show(){cout <<setiosflags(ios::fixed);    //具体显示几位小数的设置
9      cout<<"("<<setprecision(2)<<x;
10     if (y≥0){
11         cout << "+" <<setprecision(2)<<y<< "i)";
12     }
13     else{
14         cout<<setprecision(2)<<y<<"i)";}
15     }
16     Cmycomplex operator+(Cmycomplex &z)//返回值类型是Cmycomplex
17     {
18         Cmycomplex t;
19         t.x=x+z.x;
20         t.y=y+z.y;
21         return t;
22     }

```

C++

```

23     Cmycomplex operator*(Cmycomplex &z)//返回值类型是Cmycomplex
24     {
25         Cmycomplex t;
26         t.x=x*z.x-y*z.y;
27         t.y=x*z.y+z.x*y;
28         return t;
29     }
30     Cmycomplex operator-(Cmycomplex &z)//返回值类型是Cmycomplex
31     {
32         Cmycomplex t;
33         t.x=x-z.x;
34         t.y=y-z.y;
35         return t;
36     }
37     Cmycomplex operator/(Cmycomplex &z)//返回值类型是Cmycomplex
38     {
39         Cmycomplex t;
40         t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
41         t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
42         return t;
43     }
44     bool operator==(Cmycomplex &z)//返回值类型是Cmycomplex
45     {
46         if(x==z.x&&y==z.y)
47             return true;
48     }
49     bool operator!=(Cmycomplex &z)//返回值类型是Cmycomplex  //!=
50     {
51         if(x!=z.x||y!=z.y)
52             return true;
53     }
54 private:
55     double x,y;
56 };
57 //StudybarCommentBegin
58 int main()
59 {
60     Cmycomplex z1(3,4),z2(7),z3;
61     double x,y;
62     cin>>x>>y;
63     z1.Set(x,y);
64     cin>>x>>y;
65     z2.Set(x,y);
66     if(z1==z2)
67         cout<<"y\n";
68     else
69         cout<<"n\n";
70     if(z1!=z2)
71         cout<<"y\n";
72     else
73         cout<<"n\n";
74 }
75 //StudybarCommentEnd

```

复数平方根

```

1  #include<iostream>
2  #include <cmath>
3  using namespace std;
4  class Cmycomplex{
5  private:
6      double x,y;
7  public:
8      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;}
9      void Set(double xx,double yy){x=xx;y=yy;}
10     void Show(){cout<<"("<<x;
11     if (y>=0){
12         cout << "+" << y << "i)";
13     }

```

C++

```

14     else{
15         cout<<y<<"i)";}
16     }
17     Cmycomplex operator+(Cmycomplex &z)//返回值类型是Cmycomplex
18     {
19         Cmycomplex t;
20         t.x=x+z.x;
21         t.y=y+z.y;
22         return t;
23     }
24     Cmycomplex operator*(Cmycomplex &z)//返回值类型是Cmycomplex
25     {
26         Cmycomplex t;
27         t.x=x*z.x-y*z.y;
28         t.y=x*z.y+z.x*y;
29         return t;
30     }
31     Cmycomplex operator-(Cmycomplex &z)//返回值类型是Cmycomplex
32     {
33         Cmycomplex t;
34         t.x=x-z.x;
35         t.y=y-z.y;
36         return t;
37     }
38     Cmycomplex operator/(Cmycomplex &z)//返回值类型是Cmycomplex
39     {
40         Cmycomplex t;
41         t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
42         t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
43         return t;
44     }
45     Cmycomplex sqrt()//返回值类型是Cmycomplex
46     {
47         Cmycomplex t;
48         t.x=std::sqrt(x/2+std::sqrt(x*x+y*y)/2);
49         t.y=y/std::sqrt(2*x+std::sqrt(x*x+y*y)*2);//std::指明定义空间
50         if (x<0){x=-x;y=-y;}
51         return t;
52     }
53 };
54
55 //StudybarCommentBegin
56 int main()
57 {
58     Cmycomplex z1(3,4),z2(-1),z3;
59     double x,y;
60     cin>>x>>y;
61     z1.Set(x,y);
62     z3=z1.sqrt();
63     z3.Show();
64     cout<<"\n";
65     z3=z3*z2;
66     z3.Show();
67 }
68 //StudybarCommentEnd

```

复数友元函数 1

```

1  #include<iostream>
2  #include <cmath>
3  using namespace std;
4  class Cmycomplex{
5  private:
6      double x,y;
7  public:
8      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;}
9      void Set(double xx,double yy){x=xx;y=yy;}
10     void Show(){cout<<"("<<x;
11     if (y≥0){

```

C++

```

12     cout << "+" << y << "i)";
13 }
14 else{
15     cout<<y<<"i)";}
16 }
17 Cmycomplex operator+(Cmycomplex &z)//返回值类型是Cmycomplex
18 {
19     Cmycomplex t;
20     t.x=x+z.x;
21     t.y=y+z.y;
22     return t;
23 }
24 Cmycomplex operator*(Cmycomplex &z)//返回值类型是Cmycomplex
25 {
26     Cmycomplex t;
27     t.x=x*z.x-y*z.y;
28     t.y=x*z.y+z.x*y;
29     return t;
30 }
31 Cmycomplex operator-(Cmycomplex &z)//返回值类型是Cmycomplex
32 {
33     Cmycomplex t;
34     t.x=x-z.x;
35     t.y=y-z.y;
36     return t;
37 }
38 Cmycomplex operator/(Cmycomplex &z)//返回值类型是Cmycomplex
39 {
40     Cmycomplex t;
41     t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
42     t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
43     return t;
44 }
45 Cmycomplex sqrt()//返回值类型是Cmycomplex
46 {
47     Cmycomplex t;
48     t.x=std::sqrt(x/2+std::sqrt(x*x+y*y)/2);
49     t.y=y/std::sqrt(2*x+std::sqrt(x*x+y*y)*2);//std::指明定义空间
50     if (x<0){x=-x;y=-y;}
51     return t;
52 }
53 };
54
55 //StudybarCommentBegin
56 int main()
57 {
58     Cmycomplex  z1(3,4),z2(-1),z3;
59     double x,y;
60     cin>>x>>y;
61     z1.Set(x,y);
62     z3=z1.sqrt();
63     z3.Show();
64     cout<<"\n";
65     z3=z3*z2;
66     z3.Show();
67 }
68 //StudybarCommentEnd

```

复数友元函数 2

C++

```

1 //StudybarCommentBegin
2 #include <iostream>
3 #include <iomanip>
4 #include <string>
5 //StudybarCommentEnd
6 class ComplexNumber{
7 private:
8     double x,y;
9 public:

```

```

10  ComplexNumber(double xx=0,double yy=0){x=xx;y=yy;}
11  void Set(double xx,double yy){x=xx;y=yy;}
12  void Show(){
13      //具体显示几位小数的设置
14      if (x==static_cast<int>(x)&&y==0){
15          std::cout<<std::fixed<<std::setprecision(0)<<x;}
16      else {
17          std::cout<<std::fixed<<std::setprecision(2)<<x;}
18      if (y>0){
19          std::cout << "+" <<std::fixed<<std::setprecision(2)<<y<< "i";}
20      if (y<0){
21          std::cout <<std::fixed<<std::setprecision(2)<<y<< "i";}
22      }
23  ComplexNumber operator+(const ComplexNumber &z) //返回值类型是Cmycomplex
24  {
25      ComplexNumber t;
26      t.x=x+z.x;
27      t.y=y+z.y;
28      return t;}
29  ComplexNumber operator*(const ComplexNumber &z) //返回值类型是Cmycomplex
30  {
31      ComplexNumber t;
32      t.x=x*z.x-y*z.y;
33      t.y=x*z.y+z*x*y;
34      return t;}
35  ComplexNumber operator-(const ComplexNumber &z) //返回值类型是Cmycomplex
36  {
37      ComplexNumber t;
38      t.x=x-z.x;
39      t.y=y-z.y;
40      return t;}
41  ComplexNumber operator/(const ComplexNumber &z) //返回值类型是Cmycomplex
42  {
43      ComplexNumber t;
44      t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
45      t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
46      return t;}
47  //友元函数操作
48  friend std::istream &operator>>(std::istream &in, ComplexNumber &z) {
49      in >> z.x >> z.y;
50      return in;
51  }
52  friend std::ostream &operator<<(std::ostream &out, const ComplexNumber &z) {
53      if(z.x==static_cast<int>(z.x)&&z.y==0){
54          out << std::fixed << std::setprecision(0) << z.x;}
55      else{
56          out << std::fixed << std::setprecision(2) << z.x;}
57      if (z.y >0) {
58          out << "+" << std::fixed << std::setprecision(2) << z.y << "i";}
59      if (z.y<0) {
60          out << std::fixed << std::setprecision(2) << z.y << "i";}
61      return out;
62      }
63  };
64  //StudybarCommentBegin
65  void CN() { //ComplexNumber
66      ComplexNumber cn1, cn2;
67      std::cin >> cn1 >> cn2;
68      std::cout<<cn1+cn2 << std::endl;
69      std::cout<<cn1 - cn2 << std::endl;
70      std::cout<<cn1*cn2 << std::endl;
71      std::cout<<cn1 / cn2 << std::endl;
72  }
73  int main(void) {
74      CN();
75      return 0;
76  }
77  //StudybarCommentEnd

```